Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Antin, Parker B Ph.D.		
Project Title:	Calpains in Simulated Microgravity Induced Atrophy		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	pba@u.arizona.edu	**Fax:**	FY 520-626-2097
PI Organization Type:	UNIVERSITY	**Phone:**	520-626-6382
Organization Name:	University of Arizona		
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PI Web Page:	http://www.cba.arizona.edu/		
City:	Tucson	**State:**	AZ
Zip Code:	85724	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	03/01/2001	**End Date:**	06/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-MA.002.04		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Experiments will investigate the outcome of forced over expression of calpastatin in vivo. Calpastatin cDNAs driven by muscle-specific inducible promoters will be used to generate transgenic mice. Experiments are using an optimized tet on system to generate transgenic mice in which expression of calpastatin specifically in skeletal muscle cells is controlled by administration of Doxycyline. Hindlimb weighted and unweighted mice over expressing calpastatin are being subjected to detailed analyses for changes in overall muscle size, fiber diameter, nucleus/cytoplasmic ratio and protein turnover rates.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Calpains are a group of calcium activated neutral proteases found in all eukaryotic cells. Calpains target many protein constituents of the sarcomere, and compelling evidence indicates that calpain-mediated proteolysis is an early step in the turnover of myofibrils. Calpastatin is a protein that binds specifically to calpains and inhibits their function, and studies have shown that calpain activity is directly modulated by changes in calpastatin protein levels. Reducing calpain activity by increasing calpastatin levels may therefore provide an effective means for efficiently reducing or preventing muscle atrophy.
Task Progress:	No progress reported this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)